

Event-driven shifts in river and reservoir sediment sources: Cottonwood River and John Redmond Reservoir, USA

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Abstract

Reservoir sedimentation poses a critical threat to water storage capacity globally, particularly in the U.S. Great Plains where storage loss has continually declined over the past several decades. While regional management often prioritizes streambank stabilization, effectively targeting mitigation requires distinguishing between channel-derived and upland sediment sources across varying flow regimes. This study employed sediment fingerprinting to apportion streambank, cropland and grassland contributions to the Cottonwood River and the downstream John Redmond Reservoir, utilizing a multi-method sampling approach to capture different temporal scales of sediment transport. For flows contained within the river channel, time-integrated traps identified streambanks as the dominant sediment source (~50%), whereas discrete storm samples underestimated bank contributions (~35%). Statistical analysis of flow regimes and trap positioning indicates this discrepancy is not driven by vertical stratification of the sediment load or particle size differences but rather by the ability of time-integrated samplers to capture pulses of sediment transport that discrete sampling misses. In contrast, lakebed deposits in John Redmond Reservoir, which represent a longer-term record of sediment transported by large-magnitude floods, were dominated by cropland sources (~40%). The event-driven shifts in sediment provenance suggest a hydro-geomorphic threshold: during extreme, out-of-bank flows, hydrologic connectivity expands beyond the channel to the cultivated floodplain, mobilizing vast quantities of surficial cropland soil in exceedance of the river's bank-dominated internal load. Consequently, watershed management faces a dual challenge: streambank stabilization is essential to reduce the chronic sediment pulses delivered during routine within-bank flows, but soil conservation on croplands is equally critical to mitigate the large-scale sediment delivery associated with extreme out-of-bank floods.

KEYWORDS

erosion, flood events, reservoirs, Sediment fingerprinting, sedimentation, streambanks

1 | INTRODUCTION

Sediment erosion is a naturally occurring process and one sensitive to land use and climatic factors (Syvitski et al., 2022). Excess sediment erosion can degrade aquatic habitats, reduce water quality, increase water-treatment costs, threaten infrastructure stability and shorten

the useful life of reservoirs (Owens, 2020). Reservoir sedimentation and the associated loss of storage capacity, in particular, is a significant global issue (Perera et al., 2023; Wisser et al., 2013). In the United States, reservoir storage capacity over the past 30 years has continually declined due to sedimentation, resulting in estimated storage losses of 35% per capita (Randle et al., 2021). In Kansas, three of

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the state's 24 federal reservoirs are expected to become more than 50% infilled by 2030 and another eight by the turn of the century (deNoyelles & Kastens, 2016; Rahmani et al., 2018). Regionally, it is currently understood that streambank erosion contributes significantly to total watershed sediment yield and, ultimately, to sedimentation in downstream reservoirs (Juracek & Ziegler, 2009; Simon & Rinaldi, 2000).

In an effort to mitigate the harms of reservoir sedimentation, particularly from the erosion of streambanks, river restoration and streambank stabilization projects have received much attention as a potential solution (Fox et al., 2016). In Kansas, the Cottonwood River, which drains into John Redmond Reservoir, has been the source of considerable streambank stabilization efforts in the last two decades (Layzell et al., 2022). Aerial drone-based surveys show that streambank stabilization can lead to appreciable reductions in local bank erosion (Layzell et al., 2022). However, it is not certain whether stabilization projects reduce bank erosion on a spatial scale broad enough to make a significant difference in reducing sediment loads to downstream reservoirs. Thus, there is a need to apply techniques and methods that provide a more holistic, watershed-scale assessment of sediment dynamics to supplement local assessments.

Sediment fingerprinting, a modelling tool that can 'unmix' a composite sediment sample into its constituent source fractions, is a potential method for identifying source areas of erosion (Collins et al., 2017). Sediment fingerprinting works by relating the value of physical, geological or chemical characteristics (termed 'tracers') within a mixture sample, such as river sediment, to the value of these tracers in source material, such as streambank, cropland and grassland sites (Juracek & Ziegler, 2009). No single tracer is able to accurately distinguish all sources simultaneously, thus multiple tracers are needed (Davis & Fox, 2009). The type of source discrimination of interest determines the tracers that should be used (Smith & Blake, 2014). Organic tracers aid in the differentiation of the vegetative origin of sediment, such as grassland versus cropland. Radioisotopic tracers help to distinguish surficial (e.g., topsoil) versus subsurface (e.g., bank) sediment sourcing (Zhang et al., 2016). Geochemical tracers can help to distinguish sediment originating from different geologic regions of a catchment, such as Triassic versus Permian rock types (Haddadchi et al., 2013). However, sediment transport is dynamic and non-linear. Tracer signatures in suspended sediment can vary significantly as a function of discharge (Percich et al., 2022), indicating shifting sources of sediment. Consequently, there is a need to sample during multiple periods (e.g., points along a storm hydrograph) and with multiple mixture types (e.g., discrete samples, time-integrated traps and lakebed deposits) to assess potential bias in source apportionment results.

Mixing models are mathematical approaches that provide a framework for understanding the various ways sediment sources can be combined to achieve the observed sediment fingerprint in a mixture sample (Davis & Fox, 2009). Sediment fingerprinting results integrate the signal from the entire upstream drainage and thus they provide a 'global' assessment of sediment transport, as opposed to more 'local' assessments provided by aerial surveys or erosion pins (Walling, 2005). Conducting sediment fingerprinting studies with multiple sampling locations, such as repeated longitudinal sampling at points along a river, has the potential to lead to a more robust

understanding of changing sediment dynamics (Blake et al., 2018). Additionally, conducting sediment studies with multiple mixture types can provide a comparison of short-term versus long-term sediment dynamics, such as snapshots in time (via discrete storm samples) versus long-term dynamics (via flood deposits) (Tiecher et al., 2017). Comparing these mixture types also provides insight into catchment-scale connectivity, potentially distinguishing between material delivered during routine channelized flows versus high-magnitude, out-of-bank events. Once dominant sources of sediment have been identified, through these global insights, targeted sediment management strategies can be used to reduce problematic erosion sources (Sharpley et al., 2006).

The overall objectives of this project were to: (1) identify the relative contribution of streambank, cropland and grassland sediment sources to the Cottonwood River and John Redmond Reservoir; (2) evaluate the influence of sampling methodology (discrete storm grabs vs. time-integrated traps) on sediment source apportionment; and (3) assess longitudinal trends in sediment composition along a study reach subject to extensive bank stabilization. To achieve these objectives, we sampled source, river mixture and lakebed sediment during the spring and summer of 2024. Thereafter, a Bayesian mixing model was constructed to apportion river mixture and reservoir samples to the respective sources.

2 | PROCEDURES

2.1 | Study site

The Cottonwood River, located in the Flint Hills region of eastern Kansas, drains 4,928 km² (Figure 1a). The Flint Hills are characterized by erosional topography with an extensive series of *cuestas* that are controlled by successions of Permian cyclothems. Cyclothems are defined as alternating units of highly resistant cherty and flinty limestones and more erodible shales. Land use in the Cottonwood watershed comprises approximately 68% grassland, 26% cropland and 6% woodland, urban and water (Figure 1a).

The portion of the Cottonwood River assessed in this study is a low-gradient meandering river. The river is incised with typical bank heights in the study area ranging between 7 and 9 m. Despite regulation from Marion Reservoir in the headwaters of the Cottonwood River watershed, stream flows are generally still storm-generated, with high flows coinciding with the heaviest rainfall during the spring and summer months. Along the study reach, 34 stabilization projects have been constructed since 2011 (Figure 1b). The length of bank stabilized is typically on the order of 100–300 m (Layzell et al., 2022) with some larger projects stabilizing up to 700 m of streambank.

John Redmond Reservoir, constructed from 1959 to 1964, was designed for flood control, water supply and recreation (USACE, 2013). Documented sedimentation rates are almost double the original design rates (Podolak & Doyle, 2015) and have reduced storage capacity by almost 40% (Rahmani et al., 2018), thereby threatening the multi-purpose use of the reservoir. Recent mitigation efforts include storage reallocation in 2013 and dredging operations in 2016 that removed ~2,300,000 m³ of sediment at a cost of \$20,000,000 (KWO, 2025).

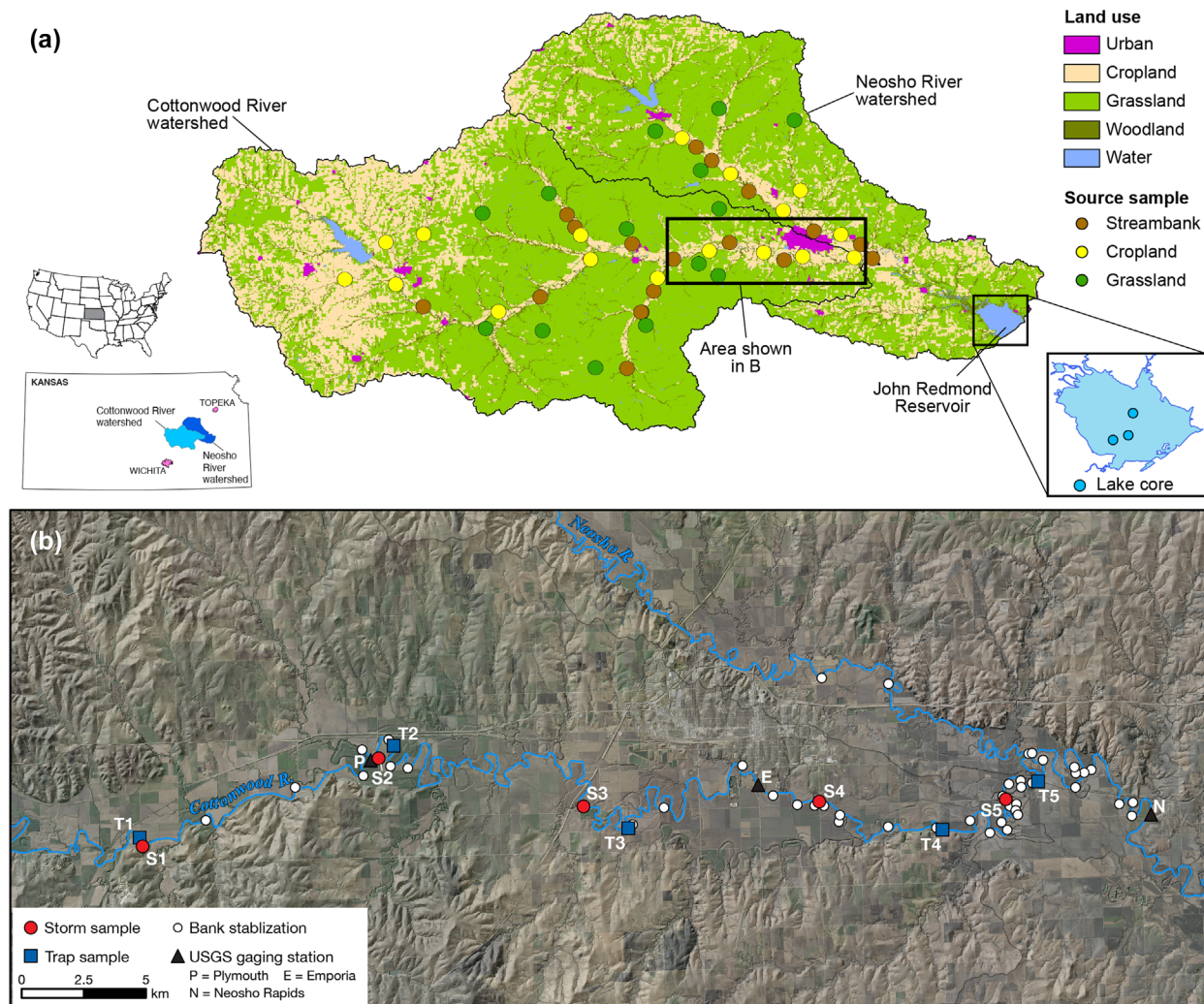


FIGURE 1 (a) Study area location showing land use and sampling sites for streambank, cropland and grassland samples in the Cottonwood and Neosho River watersheds. Inset map shows reservoir coring locations. (b) Storm and trap sampling locations along the Cottonwood River. Locations of streambank stabilization sites are also shown.

2.2 | Sample collection

Two types of sediment were collected to describe the river mixture: discrete storm (S) samples and time-integrated trap (T) samples. Each sample type was collected at five locations along the Cottonwood River, beginning upstream of the heavily stabilized corridor and ending prior to the confluence of the Cottonwood and Neosho Rivers, separated by 79 river kilometres (see Figure 1b for site locations). Sites were determined based on ease of access, such as from public right-of-way or through property owner permission. When possible, the storm and trap sampling sites were co-located. The approximate distance between adjacent sampling sites ranges from 5 to 20 km. The United States Geological Survey (USGS) operates two stream gages within the study corridor, Plymouth (07182250) and Emporia (07182260), which provide daily streamflow observations. The two gages are approximately 42 river kilometres apart.

Discrete storm samples represent instantaneous collections of suspended sediment that describe the specific transport conditions at the moment of collection. Two series of storms occurred during the study (Figure 2). The first sampling period was from April 26 to May 5, 2024, and the second from June 6 to June 15, 2024. During these periods, samples were collected from the five sampling locations

every 1 to 3 days. Stream discharge at the time of sampling ranged between 15.6 and 92.6 m³/s (stream stage 1.5–4.3 m) at Plymouth and 27.6 to 208.7 m³/s (stream stage 1.4–4.4 m) at Emporia. Samples were collected by deploying a 1-L Van Dorn sampler from a bridge and sampling suspended sediment from slightly below the water surface (Figure 3a). The sampler was emptied into one or more 5-gal buckets for transport back to the laboratory. Sampler deployment was repeated several times at a site to retrieve sufficient mass for subsequent fingerprinting analysis. Field duplicates were collected on six occasions to assess natural variability and ensure the representativeness of the sampling method.

Trap samples are time-integrated collections of sediment, which give a time-averaged view of the nature of the river sediment over the duration of the trap deployment (Phillips et al., 2000). Unlike discrete grabs, these samplers continuously accumulate sediment throughout the hydrograph, including during the rising limb, peak flow and recession, potentially capturing transient transport phases that instantaneous sampling might miss. Sediment traps were installed near the thalweg of the Cottonwood River at all five sampling locations (T1 to T5; Figure 3b). At three of the locations (T1, T3 and T5), a second trap was installed at a higher elevation (Figure S1). These paired deployments allowed for the assessment of vertical sediment heterogeneity, serving

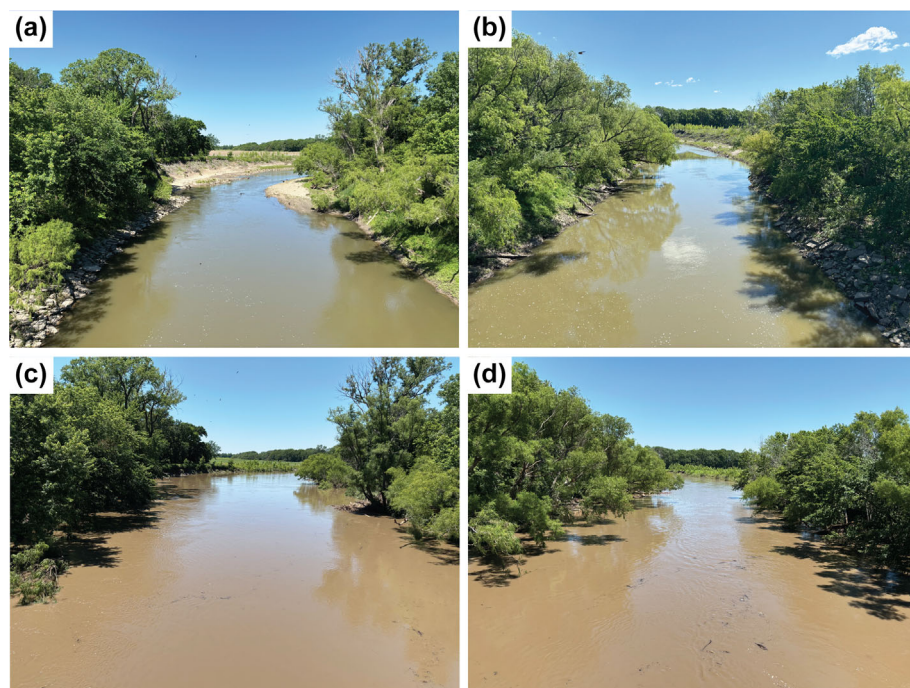


FIGURE 2 Photographs of Cottonwood River stream flow at site S4: (a) upstream, low-magnitude flow; (b) downstream, low-magnitude flow; (c) upstream, high-magnitude flow; (d) downstream, high-magnitude flow. Low flow photos taken May 10, 2024, discharge $12.9 \text{ m}^3/\text{s}$ (stream stage 1.0 m) at Emporia. High flow photos taken June 12, 2024, discharge $208.7 \text{ m}^3/\text{s}$ (stream stage 4.4 m) at Emporia.

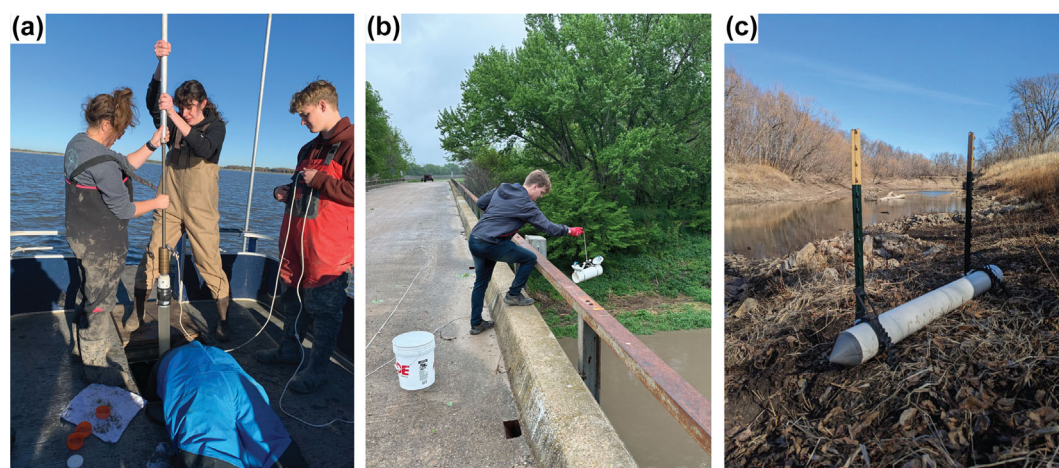


FIGURE 3 (a) Collection of lakebed samples from John Redmond Reservoir via sediment coring. (b) Deployment of 1-L Van Dorn sampler from bridge at site S2 to sample suspended sediment during a flood event. (c) Time-integrated sediment trap (lower) at site T5. The lower trap is positioned such that sediment collection only occurs when river stage exceeds $\sim 1.0 \text{ m}$.

as a proxy for field replicates to test the robustness of the integrated signal. The traps were installed such that the lower traps were submerged and infilled when river stage exceeds $\sim 1.0\text{--}1.5 \text{ m}$, and the upper traps infilled when stage exceeds $\sim 2.5\text{--}3.0 \text{ m}$. The samplers function by reducing flow velocity. They transition from a small-diameter ($\sim 0.5 \text{ cm}$), high velocity inlet to a large-diameter ($\sim 10 \text{ cm}$), low velocity pipe, promoting deposition and retention of sediment. A small diameter ($\sim 0.5 \text{ cm}$) outlet allows relatively sediment-free water to flow out of the back of the tube. The time-integrated traps were installed on April 24th and collected on an approximately monthly basis through September 4th, 2024, resulting in four different periods of analysis. Upon retrieval, trap contents were transferred into 5-gal buckets and transported to the laboratory for analysis.

Lakebed samples were collected via sediment cores at three locations in John Redmond Reservoir (Figure 1a). Sediment cores were collected with an Aquatic Research Instruments piston core with

aluminium rods to push the core tube into the lake sediment down to pre-impoundment material. All cores were extruded and sectioned at 2 cm intervals. Each section was homogenized and frozen for subsequent laboratory analyses. The upper two samples ($0\text{--}4 \text{ cm}$) from each of the three sediment cores were used to represent the reservoir mixture in the sediment fingerprinting analysis.

Source samples were collected from streambank, cropland and grassland sites (Figure 1). Cropland and grassland sites were initially selected from satellite-based land cover classification maps (Kansas Land Cover 1990, 2005, 2015 Level 1; KARS, 2024). Selected sites had consistent land use from 1990 to 2015 and current land use was confirmed in the field. Most source samples were from the Cottonwood River watershed, but a small fraction were collected from the Neosho River watershed, which also drains into John Redmond Reservoir. Source samples from both watersheds were aggregated into one dataset as they shared strong similarity. Sixteen streambanks were

sampled following standard methods, which include scraping bank material from slightly above the water line at several locations within a reach (Percich et al., 2022). Fifteen cropland samples were collected by disturbing surface sediment with a trowel, collecting several sub-samples at various locations at the same site, and aggregating these sub-samples to average out local spatial heterogeneity (Juracek & Ziegler, 2009). Thirteen grassland sites were sampled in the same manner as the cropland sites.

2.3 | Laboratory analysis

The samples were analysed for 29 potential tracers: four organics, 24 major and minor elements and a radioisotope (detailed laboratory methods are provided in Supporting Information: Methods S1). All samples were freeze-dried and sieved on a GilSonic Sonic Sifter to isolate the <63 μm fraction. For organics, the isotopic values of carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$) as well as percentages of sediment organic carbon (SOC) and nitrogen (SN) were determined at the University of Arkansas Stable Isotope Lab. The determination of the radioisotopic activity of plutonium ($^{239} + ^{240}\text{Pu}$) and semi-quantitative analysis of 24 major and minor elements (Na, Mg, Al, K, Ca, Ti, V, Cr, Fe, Co, As, Rb, Sr, Zr, Sb, Ba, La, Ce, Hf, W, Tl, Pb, Th, U) were conducted at Northern Arizona University (Ketterer et al., 2004). The distribution of all tracer values for the three mixture types (Lake, Storm and Trap) and the three sources (Bank, Cropland and Grassland) is shown in Supporting Information: Figure S2.

The size of a particle can bias the elemental concentration or radioactivity results; thus, we performed particle size analysis with a Beckman Coulter LS 13-320 laser diffraction analyser at the Kansas Geological Survey. To investigate differences in particle size distributions between sources and sinks, we performed (1) a comparative analysis across catchment sources and sediment sampling methodologies and (2) particle-size correction screening using a robust regression approach (Lacey et al., 2017) (See Supporting Information: Methods S1). Ultimately, while the mixtures ($d_{50} = 6$ to $11 \mu\text{m}$) were typically finer than the sources ($d_{50} = 13$ to $19 \mu\text{m}$; Supporting Information: Figure S3), we found that our tracer dataset is largely insensitive to particle size variations within the silt-clay fraction dominant in this system (Supporting Information: Table S1). Hence, we do not correct for size effects in the subsequent analysis.

Because the mass of sediment collected by time-integrated traps and discrete storm sampling is a function of variable field conditions, some river mixture samples had insufficient mass for all analyses. Hereinafter, we consider the results only for those samples that were analysed for all tracers.

2.4 | Statistical analysis

2.4.1 | Assessment of sampling uncertainty

Prior to tracer screening, we assessed the reproducibility of the sampling methods to ensure that observed spatiotemporal variations exceeded any potential noise present in field and laboratory conditions. We calculated the coefficient of variation (CV) for field

replicates of discrete storm samples ($n = 6$ pairs) and for paired vertical trap deployments ($n = 11$ pairs). The variability in replicate or paired samples ('noise') was compared against the total spatiotemporal variability ('signal') of the dataset for each tracer (Figure 4). This validation is necessary given the restricted spatial extent ($\sim 20 \text{ km}$ reach) and temporal clustering of samples along the hydrograph. Yet, for nearly all tracers, the environmental signal consistently exceeded the replicate noise. The ability to resolve distinct environmental signatures confirms that the sampling protocols were sufficiently robust for source apportionment.

2.4.2 | Tracer selection

Not every tracer that is analysed is ultimately useful for sediment unmixing as some tracers provide either redundant or non-informative information (Collins & Walling, 2002). We consider tracers informative if they provide discriminatory power for attributing a sediment mixture to a source. Three statistical approaches are used for this purpose: the Range Test, the Kruskal-Wallis Test and Discriminant Function Analysis (DFA). These tests identify and retain an optimal tracer set while excluding nonconservative, nondiscriminatory and redundant tracers (Lizaga et al., 2020).

The Range Test assessed the conservative behaviour of each tracer by defining a mixing envelope based on the source end members. Tracers were rejected if the median concentration of the mixture fell outside the minimum and maximum median concentrations of the combined sources. Subsequently, the Kruskal-Wallis Test was applied to determine if tracers could statistically differentiate between sources. Tracers were retained only if they exhibited significant differences ($p < 0.05$) between at least two of the three source categories (bank, cropland and grassland). Finally, a stepwise Discriminant Function Analysis (DFA) was employed to resolve the optimum composite signature. This multivariate approach iteratively entered tracers into the model based on their ability to minimize Wilks' Lambda, thereby identifying the smallest subset of variables that maximizes between-group variance relative to within-group variance. These screening steps were conducted separately for each mixture type to account for scenario-specific data constraints, resulting in distinct optimal tracer subsets (Supporting Information: Tables S2–S4).

2.4.3 | Dimensionality reduction

With informative and optimal tracers identified, we explored the potential mixing space of the three endmembers (banks, cropland and grassland) with the lakebed and the river (storm and trap) mixtures. First, we employed principal component analysis (PCA), a method for reducing the dimensionality of a variable space, to visually describe the mixing process in two dimensions. To aid in visually assessing similarities and differences between sediment sources, confidence ellipses were generated with a one standard deviation bound. We conducted a permutational multivariate analysis of variance test (perMANOVA) to identify if the centroid and dispersion are significantly different ($\alpha = 0.05$) between source groups in the PCA space (Anderson, 2017).

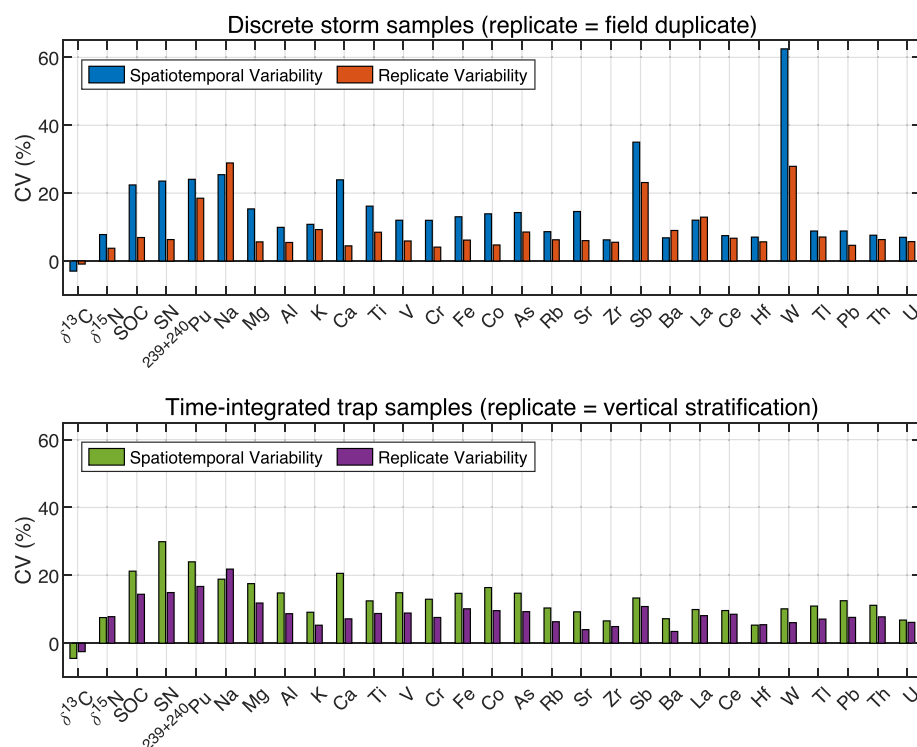


FIGURE 4 Assessment of variability for all potential tracers analysed, as expressed by the coefficient of variation (CV). The grouped bar charts compare spatiotemporal variability (signal) against replicate variability (noise) for all 29 tracers. The top row shows an analysis of discrete storm samples ($n = 33$), where spatiotemporal variability (blue bars) represents the variance observed in tracer values across different monitoring sites and sampling events. Replicate variability (orange bars) is derived from co-located field duplicates ($n = 6$). The bottom row shows an analysis of time-integrated trap samples, where spatiotemporal variability (green bars) is confined to traps positioned nearer the streambed (Lower; $n = 11$) to isolate longitudinal and temporal trends from vertical stratification effects. Replicate variability (purple bars) represents the vertical stratification observed between paired Lower and Upper traps ($n = 10$).

2.5 | Mixing model

2.5.1 | Model formulation

Sediment fingerprinting models can take many forms, from fully frequentist to fully Bayesian (Cooper et al., 2014). Irrespective of the modelling framework selected, both approaches typically solve the following mass balances for sediment (1) and each tracer (2) as:

$$\sum_{i=1}^n \pi_i = 1, \pi_i \geq 0 \text{ for all } i = 1, 2, \dots, n \quad (1)$$

and

$$\sum_{i=1}^n \pi_i S_{ij} = M_j, \text{ for } j = 1, 2, \dots, m \quad (2)$$

where π_i is the fractional contribution from the i -th source, n is the number of sediment sources, m is the number of tracers, S_{ij} is the j -th tracer concentration of the i -th source and M_j is the concentration of the mixture for the j -th tracer. Constraints include $0 \leq \pi_i \leq 1$ and the sum of contributions equaling unity.

We applied a Bayesian mixing model, solved via Sampling-Importance-Resampling, independently to each mixture sample. First, distributions for the source tracer data (S_{ij}) were generated by drawing 10^6 random samples from multivariate normal distributions defined by the means and covariances of the source groups, thereby explicitly accounting for source variability and correlation. Simultaneously, a corresponding set of source proportions (π_i) was drawn from a chemically feasible space using a uniform Dirichlet distribution. These proposed source compositions and proportions were combined to predict candidate mixture concentrations. The likelihood of each candidate solution was evaluated by determining the probability density of the predicted mixture given the observed mixture and a defined

observation error. This likelihood function utilized a multivariate normal distribution where the covariance matrix, Σ_{obs} , was defined as a diagonal matrix with standard deviations corresponding to 5% of the observed mixture concentration for each tracer, representing measurement and model uncertainty. An important resampling algorithm was used to generate the final posterior distributions (Erhardt & Bedrick, 2013).

2.5.2 | Posterior analysis and testing

The model's posterior distributions were analysed at two spatial scales to address distinct research objectives. First, to characterize catchment-scale sediment sourcing, we performed posterior averaging on the observation-level results. Individual posterior probability density functions (PDFs) were generated from 1,000 draws for each sample. These were then aggregated by mixture type (Lake, Storm and Trap) to produce pooled posterior distributions representing the dominant sources for each hydrologic regime (Supporting Information: Figure S4). Second, to assess fine-scale longitudinal trends, we aggregated the observation-level results for storm and trap samples by river kilometre. Linear regression was then used to test the statistical significance of changes in mean bank contribution along the stabilized study reach.

To investigate potential mechanisms driving source apportionment differences between sampling methodologies (i.e., discrete vs. integrated), we performed a targeted statistical comparison. Trap results were stratified by flow regime (High vs. Low) and vertical position (Upper vs. Lower) and compared against discrete storm samples. Differences between these sub-groups were evaluated using non-parametric Wilcoxon rank-sum tests ($\alpha = 0.05$). Finally, to validate model robustness, we performed an inter-model assessment by comparing the Bayesian results against two frequentist approaches: constrained least-squares regression (Collins et al., 2020) and Latin Hypercube Sampling (Lizaga et al., 2020).

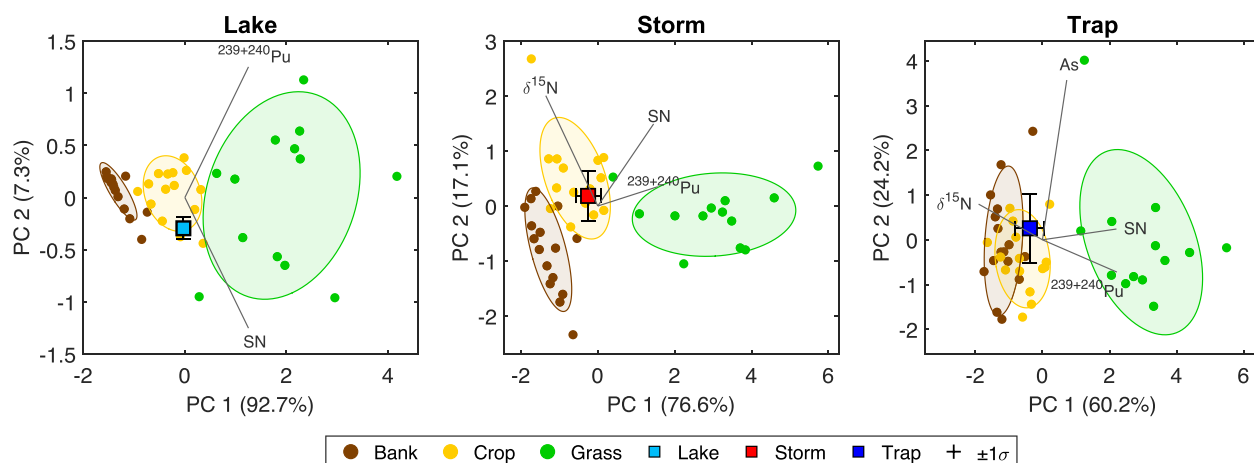


FIGURE 5 Principal component analysis (PCA) of optimal tracers for Lake, Storm and Trap sediment mixtures. PCA biplots show the multivariate structure of sediment source and mixture samples for each mixture type using the optimal tracer subsets identified by the tracer selection procedure. Panels correspond to the Lake, Storm and Trap models (left to right). Optimal tracers for each mixture type are shown as loadings. Source samples (Bank, Cropland and Grassland) are shown as individual points with shaded ellipses representing the 68% confidence region of each source group. Mixture samples are summarized by their centroid in PCA space, with error bars indicating ± 1 standard deviation along each principal component. Axis labels report the percentage of total variance explained by each principal component.

3 | RESULTS AND DISCUSSION

3.1 | River, reservoir and source sediment characteristics

The optimal tracer set varied for different mixture types: Lake (SN and $^{239} + ^{240}\text{Pu}$), Storm (SN, $\delta^{15}\text{N}$ and $^{239} + ^{240}\text{Pu}$) and Trap (SN, $\delta^{15}\text{N}$, As and $^{239} + ^{240}\text{Pu}$) (Supporting Information: Figure S5). For SN and $^{239} + ^{240}\text{Pu}$, grassland had the greatest values, followed by cropland and then bank, highlighting a gradient from surface to subsurface sediments. For $\delta^{15}\text{N}$, cropland had the greatest values, followed by bank and grassland, reflecting the effects of nitrogen inputs and cycling. Lastly, As, which was only informative for trap samples, had the greatest values in grassland, followed by bank and then cropland. Higher As levels in grassland sediments are likely related to the underlying shallow bedrock, which is comprised of marine limestones and shales.

The principal component analyses of the optimal tracer sets are shown in Figure 5. The first two principal components (PCs) explain a considerable amount of the variance (totalling 100% for lake, 93.7% for storm and 84.4% for trap) in the dataset. PC 1 appears to distinguish relatively undisturbed, organic-rich surface sites (grassland) in the positive PC 1 domain from relatively organic-poor sites (banks and cropland) in the negative PC 1 domain. The domain of PC 2, particularly for storm samples, appears to further distinguish organic-poor sediment into subsurface (bank) and disturbed surface (cropland) domains. The distributions of the three sediment source clusters are significantly different from one another as determined by a perMANOVA test ($p < 0.05$). In the PCA space, Lake, Storm and Trap samples tend to be positioned further from the grassland source and closer to the cropland and bank sources.

3.2 | Mixing model results

The models' posterior distributions were analysed at two scales: (1) a broad-scale assessment of sediment provenance for the Cottonwood

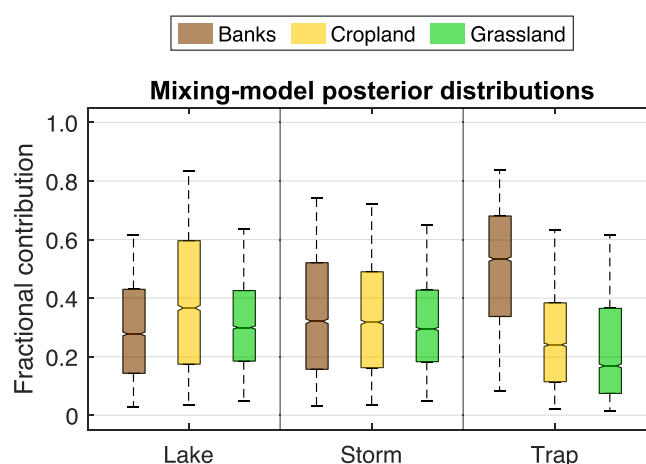


FIGURE 6 Posterior distributions of source contributions for Lake, Storm and Trap sediment mixtures. Boxplots summarize posterior distributions of fractional source contributions (Banks, Cropland and Grassland) estimated from the Bayesian mixing model for Lake, Storm and Trap sediment mixtures. For each mixture type, posterior draws are grouped by source and displayed side-by-side, with mixture types separated by vertical dividers. Boxes represent the interquartile range (25th–75th percentiles) with notches indicating the median, while dashed whiskers extend to the 5th and 95th percentiles of the posterior distributions.

River and John Redmond Reservoir and (2) a fine-scale longitudinal assessment of sediment composition at different times and locations along the study river reach.

3.2.1 | Broad-scale assessment

Figure 6 shows the respective pooled posterior distributions for lakebed, discrete storm and time-integrated trap mixtures. For lakebed samples, the mean bank, cropland and grassland fractions are 0.29, 0.40 and 0.31, respectively. Hence, cropland sources are the largest

single contributor to reservoir sediment. For storm samples, these fractions are 0.35, 0.34 and 0.31, respectively. For trap samples, they are 0.50, 0.27 and 0.23, respectively. Overall, there is a gradient from upland (grassland and cropland) sourcing for lakebed deposits to in-channel (bank) sourcing for time-integrated traps. A comparison of our Bayesian mixing model results to those of two frequentist methods (least-squares and Latin Hypercube Sampling) yielded similar results (Supporting Information: Figure S6).

There is a considerable difference in the estimation of source contributions by discrete storm samples compared to time-integrated traps, most notably for bank sources. These results contrast with those of prior studies that have demonstrated that discrete (storm) and time-integrated (traps) samples have consistent physical, chemical and biomarker properties as well as consistent sediment source apportionment results (e.g., Phillips et al., 2000; Upadhyay et al., 2024; Williamson et al., 2025). Considering these findings, the observed differences in source contributions between our storm and trap samples are unexpected, and we explore potential drivers in the subsequent paragraphs.

3.2.2 | Fine-scale assessment

Figure 7 shows the spatiotemporal dynamics of sediment sourcing for storm and trap mixtures at their specific sampling sites and times. Mixing model results are shown only for sites and times when sufficient mass was collected for laboratory analyses. We visualize the results on a split-axis hydrograph to qualitatively link shifts in sediment sources to specific flow regimes and hydrologic events (e.g., rising vs. falling limb) as well as sampling periods.

Discrete sample sets were collected six times during Storm 1. The first two sets of samples were collected when the flow stage was between 1.5 and 2.4 m and have notable variability in their source fractions. The third sample set was collected at a peak hydrograph stage of 4.3 m and consistently had a dominant bank fraction across all locations (0.46 to 0.65). The subsequent three sampling sets for Storm 1, sampled between stages of 1.9 and 2.7 m, show variable sediment sourcing during the recession of the event. In general, the discrete storm samples show considerable variability across sampling times and locations, indicating dynamic sourcing and mobilization of variable sediment sources.

During Storm 2, three discrete sample sets were collected. The first sample set was collected at a stage of 3.6 m, shortly after the hydrograph peak, and generally had consistent bank fractions of ~ 0.35 . In contrast, the second and third sample sets, taken at stages of 3.5 and 2.7 m, respectively, but further along the falling limb of a second peak in the hydrograph, displayed a lower bank fraction of ~ 0.28 , with greater contribution from croplands. To summarize, the qualitative assessment of sediment sourcing from discrete samples during Storms 1 and 2, bank fractions tend to be the largest during high stages, whereas sourcing becomes more variable at lower stages. We subsequently test this inference by considering time-integrated trap results.

Traps deployed during Periods 1 and 3 were concurrent with Storms 1 and 2, respectively, and experienced flood events with stages greater than 3 m (Figure 7). For Storm 1, the bank fraction within the time-integrated traps varied from 0.34 (upstream) to 0.58 (downstream). These mixing fractions are within the variability

observed in the discrete-storm samples for Storm 1. However, for Storm 2, the time-integrated trap samples consistently show higher bank fraction contributions (> 0.61) than the discrete storm samples (< 0.43). Notably, we were unable to collect discrete samples during the peak flows of Storm 2. These peak flows likely carried the sediment that filled the time-integrated traps, given that most sediment transport occurs during a few, extreme events.

To explore potential mechanisms for why the time-integrated traps predict greater bank fraction contributions than the discrete storm samples, we conduct several experiments to assess:

1. the model's ability to reproduce source contributions and overcome natural variability (or 'noise') in tracer signals,
2. the effect of particle size variation between discrete storm samples and time-integrated trap samples,
3. the effect of the vertical position of the time-integrated trap relative to the discrete storm samples, and
4. the effect of flow regime (high flow vs. low flow) on sediment sourcing within traps.

Model reproducibility assessment

Variation in bank fraction predictions for storm and trap samples could arise from noise in our field and laboratory methodology. To validate the reproducibility of our mixing model results, we assess the difference between replicate storm field samples and between upper and lower trap pairs (Figure 8). Results show that our model can reproduce a consistent apportionment of the bank fraction. The mean absolute difference between field replicates of discrete storm samples was 8.1%. The mean absolute difference between paired time-integrated traps was 1.3%. The paired traps were installed at approximately 1.0 to 1.5 m stage (Lower) and 2.5 to 3.0 m stage (Higher). Together, these results demonstrate the ability of the model to reliably reproduce results when tested on field replicates or vertical trap pairs.

Particle size assessment

Particle size effects could bias mixing model results, as larger particles may be captured by the traps, which are positioned near the bed, compared to the discrete storm samples, which are sampled from just below the water surface. We find significant differences in the bank fraction contribution between storm and trap samples ($p < 0.05$), but no significant differences in median particle size between the sampling methodologies ($p = 0.06$) (Figure 9). Nonetheless, the median diameter of the storm samples ($d_{50} = 8.95 \mu\text{m}$) is smaller than that of the trap samples ($d_{50} = 10.39 \mu\text{m}$). However, our earlier analysis showed that tracer values for the optimal tracer set are relatively insensitive to particle size effects (See Methods Section 2.3, Supporting Information: Methods S1, Tables S1 and S2). Thus, the slight insignificant differences in particle size between storm and trap samples are unlikely to explain the differences in bank fraction estimated by the two methods.

Vertical stratification assessment

Vertical sediment stratification effects could cause differences in sediment sourcing, as larger particles may travel closer to the bed, whereas smaller particles are transported closer to the water surface. This potential vertical stratification of particle classes could be

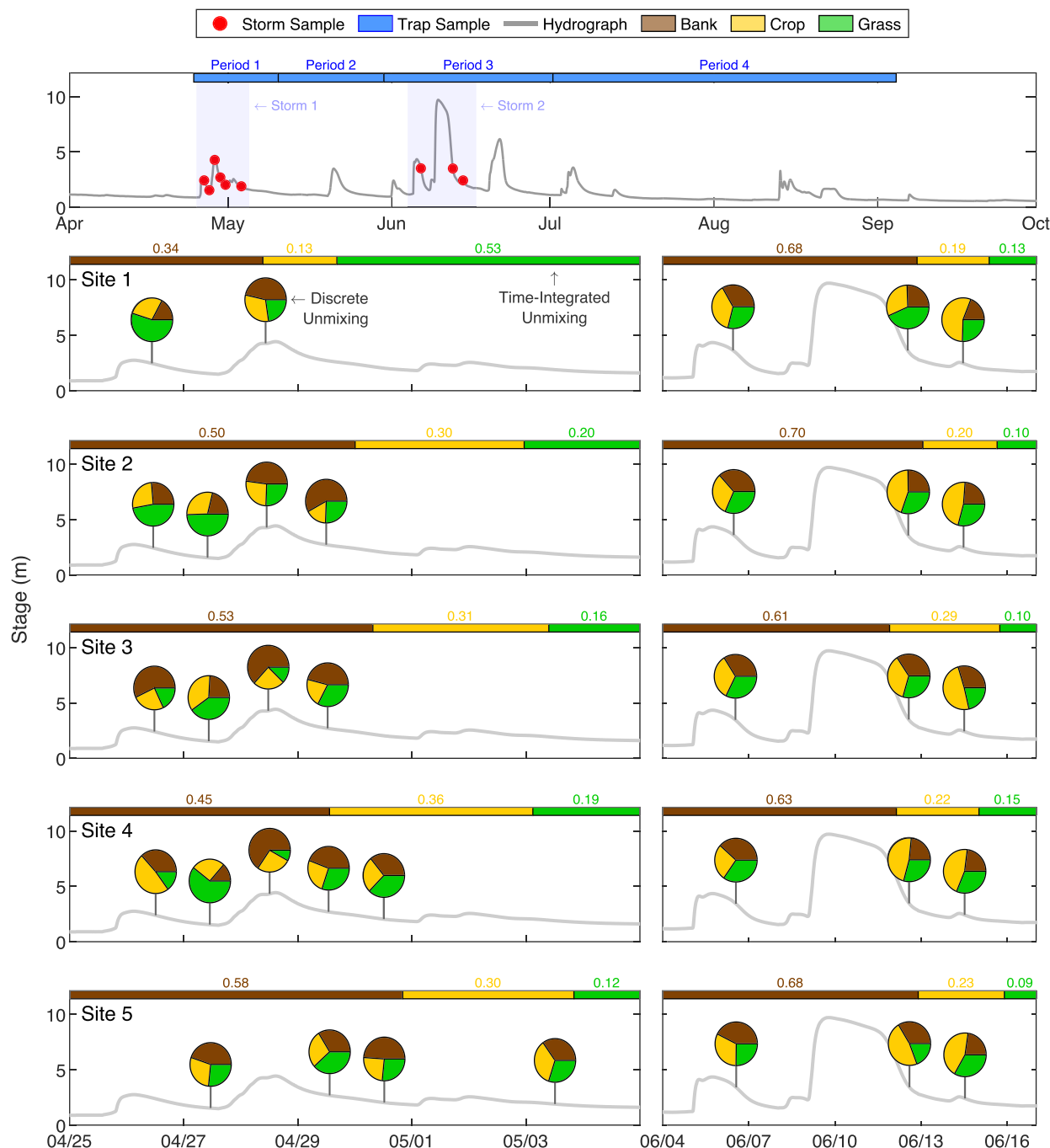


FIGURE 7 Spatiotemporal comparison of sediment source apportionment using discrete storm sampling versus time-integrated sediment traps. The top panel shows the hydrograph stage (grey line) during the study period. Red circles indicate timing of discrete storm samples. Blue bars at the top denote the deployment duration for four consecutive time-integrated trap periods. Shaded vertical bands indicate the two specific storm events. The lower panels display two zoomed time windows (left and right columns) for each of five sampling locations (rows). Pie charts represent the source contribution for discrete storm samples at specific hydrograph points. Pie charts are shown only for discrete samples with enough mass for all laboratory analyses. Horizontal ribbons at the top of each panel display the time-integrated source mixture derived from sediment traps deployed during that specific event window (Period 1 for Storm 1; Period 3 for Storm 2). Fractions above the ribbons indicate the specific source fractions for the time-integrated samples.

associated with sediment originating from different sources. Our paired sediment trap design shows no significant difference in particle size or predicted bank fraction between traps at the Lower position (~ 1.0 – 1.5 m stage) and those at the Upper position (~ 2.5 – 3.0 m stage) (Figure 9). The lack of significant differences suggests that there is complete vertical mixing and that the same sediment is filling the paired traps. Notably, there remains a significant difference between storm samples and the upper traps ($p < 0.05$). The discrete storm

samples were collected at approximately the same stage (mean = 2.70 ± 0.89 m; $n = 39$) as the upper traps (2.5 – 3.0 m). Thus, despite the upper trap and discrete samplers collecting material at approximately the same elevation, the material filling the trap contains a significantly greater fraction of bank material. Therefore, a mechanism other than noise in the environmental signal, particle size differences or vertical stratification of sediment is driving the differences in simulated bank fraction.

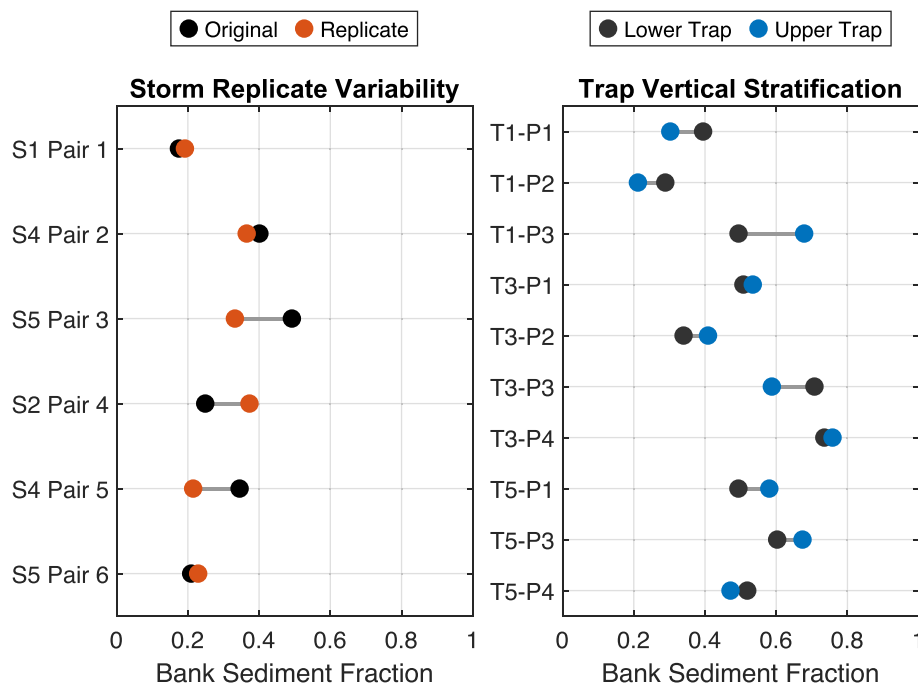


FIGURE 8 Analysis of sampling uncertainty and vertical heterogeneity in sediment source apportionment results. (Left) Assessment of methodological reproducibility using discrete storm field replicates. Dumbbell plot illustrating the deviation in modelled bank sediment contribution between original samples (black circles) and co-located field replicates (orange circles). Scatter plot comparing replicate pairs with scatter representing field sampling variability. (Right) Assessment of vertical stratification using paired time-integrated sediment traps. Comparison of bank sediment fractions in traps deployed near the streambed ('Lower,' dark grey circles) vs. further up the bank ('Upper,' blue circles) during identical monitoring periods.

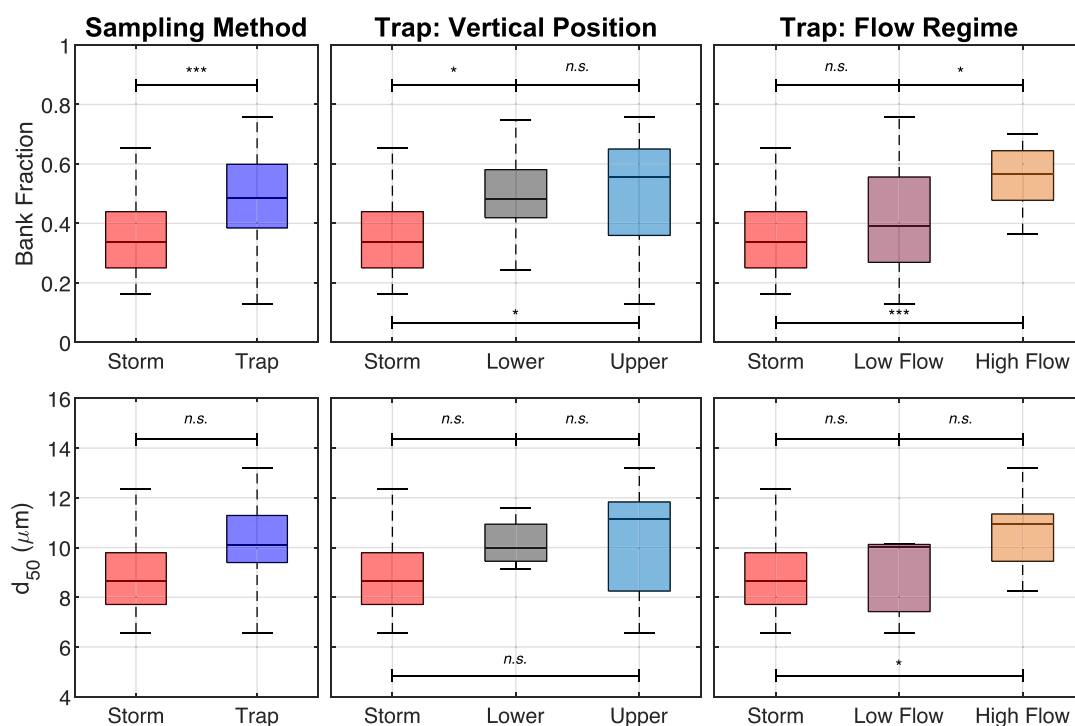


FIGURE 9 Evaluation of sampling method, vertical stratification and flow regime on sediment characteristics. The top row shows modelled bank sediment contribution, while the bottom row shows the median particle size (d_{50}). The first column shows an assessment of sampling method, contrasting instantaneous discrete storm samples ($n = 39$) against time-integrated trap samples ($n = 29$). The middle columns show an assessment of vertical stratification, comparing trap samples from paired near-bed (Lower; $n = 11$) versus upper-bank (Upper; $n = 11$) deployments. The right column shows an assessment of flow regime, comparing trap samples collected during high-flow periods (Periods 1 & 3; $n = 16$) versus low-flow periods (Periods 2 & 4; $n = 13$). Brackets indicate results of independent Wilcoxon tests with statistical significance (p -values) thresholds indicated by asterisks: *** (0.0005), ** (0.005), * (0.05) and *n.s.* (> 0.05).

Flow regime assessment

Flow intensity can amplify the degree of hydrologic connectivity of certain sediment sources. To this end, storm samples capture a discrete moment of connectivity and sediment transport, whereas traps integrate the sediment record over their period of deployment. Trap periods 1 and 3, which align with the sampling of discrete storm

samples (Figure 7), were characterized by relatively higher flow. In contrast, Periods 2 and 4, which were sampled by time-integrated traps but not discrete storm samplers, have lower flows. Therefore, we stratified the trap samples into 'High Flow' (Periods 1 and 3) vs. 'Low Flow' (Periods 2 and 4). If bank contributions increase with flow stage, we would expect bank fractions to be higher during 'High

Flow' periods relative to 'Low Flow' periods. Indeed, we see significantly higher bank fraction in 'High Flow' traps than 'Low Flow' traps (Figure 9).

Interestingly, the 'High Flow' traps also have a significantly higher bank fraction than the discrete storm samples collected during the same period. The likely reason for this discrepancy is that the time-integrated traps infilled with sediment during periods of higher flow that were missed in discrete sampling (Figure 7). The infilling of the trap during deployment does not occur uniformly, but is rather likely weighted by the sediment load, which is the product of streamflow and concentration. Thus, the time-integrated trap is likely most representative of sediment transported near the peak of stream discharge. During Storm 1, a set of discrete samples was collected at the hydrograph peak at four of the five locations (Figure 7), and all four samples showed a plurality or majority of bank-derived material. On the other hand, during Storm 2, the discrete samples missed both peaks, particularly the second peak, which reached a stage of 9 m. Therefore, the discrete samples are not representative of the bulk of sediment transported during this period. In sum, bank fraction within the time-integrated traps appears to increase with stream discharge, likely the result of enhanced bank erosion when the streamflow is largely contained within the channel (out-of-bank stage: 9.75 m), which was the case for all storms that occurred during this period (max stage: 9.70 m; Supporting Information: Figure S7).

In sum, our assessment of the observed differences in source contributions between our storm and trap samples indicates that the most likely reason for the inconsistency is the timing of sample collection. We do not have complete coverage of the hydrograph with our discrete sampling and may have missed key time-variant sediment mobilization periods, particularly from bank sources. By integrating sediment flux over the entire event, trap samplers provide a more robust estimate of sediment provenance.

3.3 | Bank erosion in the stabilized corridor

Figure 10 shows the longitudinal variation in fractional sediment source contribution along the study reach, which has 34 streambank stabilization projects. Results indicate that the fraction of suspended sediment sourced from bank sediment tends to increase downstream. For discrete storm samples, the mean bank fraction is 0.276 at the upstream-most site (S1) and 0.349 at the downstream-most site (S5), an increase of 7.3% ($p = 0.254$; Supporting Information, Figure S8). For time-integrated trap samples, the mean bank fraction is 0.406 at the upstream-most site (T1) and 0.497 at the downstream-most site (T5), an increase of 9.1% ($p = 0.311$; Supporting Information, Figure S8). While the high variability inherent in natural sediment transport precluded statistical significance ($p > 0.05$), the consistency of this increasing trend across both sampling methodologies suggests a progressive downstream increase in bank-derived material, even within a stabilized corridor. We note that mean stream discharge during the study period increases by 18% between the two gaging stations. This increase in discharge likely generates greater shear stress across the channel boundary, promoting the recruitment of bank material, particularly failed bank material at the bank toe, relative to upland sources as the river flows downstream.

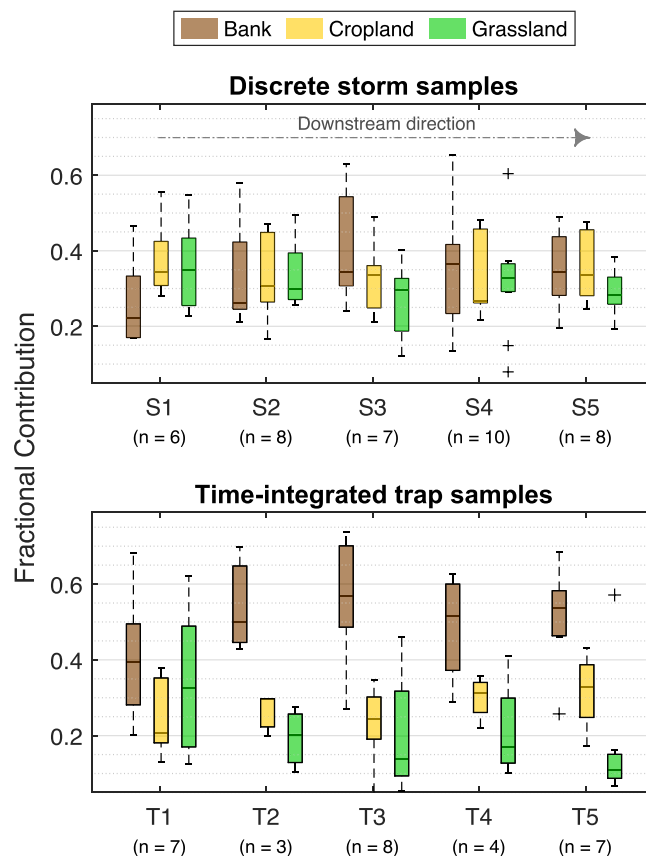


FIGURE 10 Source-contribution distributions for storm and trap samples, resolved by sampling site and ordered longitudinally (left to right indicates downstream direction). Boxplots represent the distribution of posterior means of individual discrete storms (top, $n = 39$) and time-integrated trap (bottom, $n = 29$) samples. Boxes represent the interquartile range with medians shown as horizontal lines; points indicate outliers. Sample sizes (n) for each site are reported beneath the x-axis labels. Results are derived from the Bayesian mixing model and are plotted as posterior means for each physical sample at a location.

3.4 | Long-term deposition in the reservoir

One major finding of our results is that lakebed sediment in John Redmond Reservoir is most prominently derived from cropland sources (Figure 6), in contrast to the current understanding that streambanks are the dominant contributor. These findings have implications for understanding changes in sediment sources in the Cottonwood River during different flow regimes. Overall, our results can enable, support and track targeted management of erosion sources.

During typical sediment transport conditions in the Cottonwood River, as inferred from results of the time-integrated traps, 50% of the sediment originates from bank erosion, with the remaining derived from upland sources (27% cropland, 23% grassland). Downstream reservoir sediment, however, is composed of only 29% bank material compared to 71% from upland sources (40% cropland, 31% grassland). The increase in cropland contributions to the reservoir can be explained by considering the magnitude and frequency of flood events. For the Cottonwood River, bankfull conditions represent the point where flow transitions from in-channel flow to out-of-bank flow. We suggest that high magnitude, low frequency out-

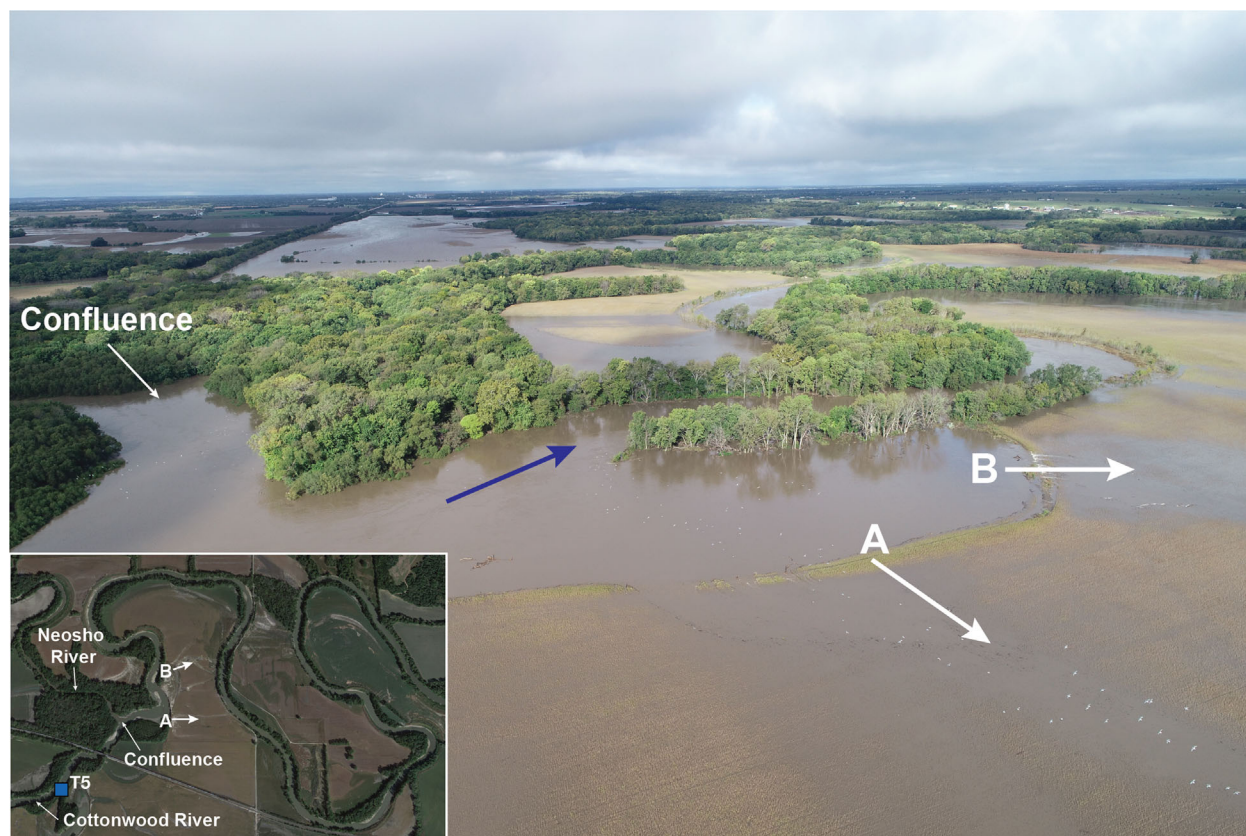


FIGURE 11 Flooding on the Cottonwood and Neosho Rivers (October 10, 2018). Maximum stage recorded during this event at the Neosho Rapids gage, approximately 14.5 river km downstream of the confluence, was 8.2 m, exceeding the bankfull height of 6.7 m. Out-of-bank flow across cropland (white arrows) and in-channel flow (blue arrow, pointing downstream) are shown. Chute cutoff channels, which short-circuit the meander bend and create erosional pathways across the floodplain, can be seen along the A and B arrows in the birds-eye-view inset.

of-bank events result in erosion and transport of cropland sediment along the heavily cultivated valley floor of the Cottonwood River. In particular, cropland sediment is likely entrained from floodplain chutes that cut across meander bends and create erosional pathways across floodplain surfaces (Figure 11).

Studies have shown that disproportionally large sediment loads are transported and delivered to downstream lakes and reservoirs during infrequent, high-magnitude flood events (Foster et al., 2012; Wolman & Miller, 1960), including the Cottonwood River draining into John Redmond Reservoir (Kramer et al., 2021). For example, Kramer et al. (2021) note that from 2016 to 2019 at the Plymouth gage, 48% of the total suspended-sediment load was transported by flood events greater than bankfull. At the Plymouth gage, bankfull flows have stages greater than 9.75 m and occupy less than 1% exceedance probability (Supporting Information: Figure S7). The percentage of total suspended-sediment load transported by out-of-bank flood events increases downstream to 78% at the Neosho Rapids gage (2010–2019) located approximately 14.5 river km below the confluence of the Cottonwood and Neosho Rivers (Kramer et al., 2021). Hence, our lakebed samples are most representative of material transported via large, infrequent out-of-bank floods. In contrast, the storm and trap samples, which collected sediment during smaller, more frequent flow events, represent transport of sediment contained within the channel, i.e., below out-of-bank stage (Supporting Information: Figure S7). Therefore, storm and trap samples, which have higher contributions from bank sources relative to lakebed samples, reflect a dominance of streambank erosion processes. Because frequent bank slumping via

mass wasting processes has been documented along the Cottonwood River (Layzell et al., 2022), we suggest that failed bank material at the bank toe is the primary source of sediment transported during the more frequent, in-channel flows captured by the discrete storm samples and time-integrated traps.

3.5 | Implications of results for management

Our results indicate that lakebed sediment in John Redmond Reservoir is most representative of material transported during large, infrequent flood events and that this sediment is primarily derived from cropland sources, not streambanks. Discrete sampling of moderately sized storm events that occur multiple times per year indicates a drop in the relative proportion of cropland sediment and an increase in bank material compared to lakebed samples. Time-integrated traps, which collected sediment most representative of periods of high transport contained within the channel, had the lowest cropland and greatest bank contributions. The shift from in-channel bank-dominated sourcing, for time-integrated traps, to cropland-dominated sediment sourcing, for lakebed material, reflects the enhanced connectivity of upland sources, particularly cropland, during high-magnitude, out-of-bank floods. Methodologically, this study demonstrates that relying on a single mixture type or sampling modality can lead to biased management recommendations. Discrete sampling alone would have underestimated bank erosion, while trap sampling alone would have missed the substantial cropland inputs during

floods. Effective watershed management requires a multi-method approach that captures both the chronic (in-channel) and acute (flood-plain) sediment loads.

Although much of the focus on reducing erosion in the Cottonwood watershed upstream of John Redmond Reservoir has been directed toward stabilizing streambanks, we show that even complete stabilization of the study reach would only partially address the excess sediment load issue. Because we show the potential for upland sources, particularly croplands, to contribute substantial sediment, targeted management should include not only streambank stabilization, but also practices aimed at reducing cropland erosion, such as cover crops, conservation tillage, riparian buffers, streambank fencing, improved irrigation efficiency and floodplain erosion control structures (Sharpley et al., 2006). The results and inferences provided by sediment fingerprinting provide an additional point of evidence in a broader set of approaches, which include aerial surveys, hydrologic modelling and long-term monitoring (Layzell et al., 2022, 2024).

3.6 | Limitations and future work

While sediment fingerprinting is a valuable approach for understanding and quantifying sources of erosion in a watershed (Walling, 2005), the technique is not without limitations. First, mixing models are most well-suited for providing broad-stroke insights into sediment provenance. Applying these at fine temporal scales is possible, but must be handled with care as tracer values can have a high degree of temporal variability. This occurs because (1) not every source (e.g., cropland field) has the exact same properties, (2) spatial heterogeneity in rainfall and runoff processes causes some fields to be connected while others are not and (3) transformation of tracers during transport can impact predictions. We attempt to mitigate these limitations by sampling at least a dozen locations for each source, distributed evenly across the watershed. Additional samples from sources, as well as river mixtures, can reduce these uncertainties.

Several avenues exist to improve model confidence, such as (1) collecting additional mixture samples, (2) including more informative tracers in the modelling process and (3) assessing change in sediment sourcing along the rising and falling limb of storms. Although data pertaining to this latter question were collected as part of this study, we determined that more discrete field sampling is needed across several storm hydrographs in order to accurately elucidate this process. This is a potential and important avenue of future work, as our results indicate inconsistency between storm and trap source apportionment. Sampling the rising and falling limbs of flood hydrographs would allow us to better elucidate sediment mobilization periods. Lastly, although we observed a net increase in bank-derived sediment downstream (approximately 7 to 9%) through the stabilized study reach, without a counterfactual scenario, we cannot determine how effective current stabilization projects truly are. A counterfactual scenario would be one that estimates what the bank fraction increase would be in the absence of the streambank stabilization projects that are currently in place, such as a sediment tracing project conducted prior to the installation of any erosion control structures. This is a crucial source of information for determining the efficacy of stabilization practices and highlights the importance of collecting adequate and representative baseline data before the implementation of stabilization practices.

4 | CONCLUSIONS

This study demonstrates that sediment source apportionment is highly dependent on both the sampling methodology and the hydrologic regime. By employing a multi-method approach—contrasting instantaneous discrete storm samples, time-integrated traps and reservoir deposits—we revealed a decoupling between the in-channel sediment load and the flood load delivered to the reservoir during extreme events.

Methodologically, we found a divergence in source attribution between sampling techniques. Discrete storm samples consistently underestimated the contribution of streambanks compared to time-integrated traps deployed during the same events. Statistical analysis ruled out particle size bias or vertical stratification as drivers of this discrepancy. Instead, we conclude that time-integrated samplers captured high-intensity pulses of bank-derived sediment that instantaneous discrete sampling missed. Consequently, reliance on discrete ‘snapshot’ sampling may lead to a misrepresentation of dominant sediment sourcing.

Hydrologically, the study identified a critical geomorphic threshold. While the in-channel load (captured by traps) is dominated by bank sediment (50%), the long-term depositional record in John Redmond Reservoir is dominated by cropland sediment (40%). This inversion indicates that during infrequent, high-magnitude out-of-bank floods, the river connects with its floodplain, mobilizing vast quantities of surficial cropland soil that overwhelm the in-channel signal. Thus, the reservoir records the signature of extreme events, while the river tracks the signature of routine flows.

Longitudinally, we observed a consistent downstream increase in the bank sediment fraction (7–9%) across the 79 km study reach. We attribute this to the progressive erosion of bank material driven by increasing discharge and shear stress, despite the presence of extensive bank stabilization projects. However, without a pre-stabilization counterfactual, the absolute efficacy of these structures remains difficult to quantify.

Ultimately, these findings indicate that management strategies should not solely target streambank stabilization but must integrate measures addressing upland erosion, particularly from cropland areas. Although sediment fingerprinting offers robust, global-scale assessments of sediment provenance, caution is necessary when interpreting results at fine spatial and temporal scales due to inherent variability in tracer characteristics and sediment transport dynamics. Future monitoring efforts should prioritize multi-method sampling strategies to ensure that both the baseline load and the event-driven pulses are accurately characterized.

AUTHOR CONTRIBUTIONS

Anthony L. Layzell: Conceptualization; funding acquisition; methodology; investigation; writing—original draft.

Cayden E. Lloyd: Investigation.

Abigail Percich: Investigation.

Ted D. Harris: Funding acquisition; investigation.

Michael E. Ketterer: Methodology; investigation.

Admin Husic: Conceptualization; funding acquisition; methodology; investigation; writing—original draft.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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